



1424 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Cycle: 1, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Jupiter Trigger - Epoch Two				
	10	JupiterMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(1) JUPITER-TARGET
	11	MIRI Background	MIRI Medium Resolution Spectroscopy	(2) JUPITER-BACKGROUND
	12	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(6) JUPITER-TARGET-NIRSPEC
	13	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(7) JUPITER-TARGET-LOC2-NIRSPEC
Saturn Trigger - Epoch One				
	5	SaturnMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(3) SATURN-TARGET
	6	MIRI Background	MIRI Medium Resolution Spectroscopy	(4) SATURN-BACKGROUND

Folder	Observation	Label	Observing Template	Science Target
	8	SaturnNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(8) SATURN-TARGET-NIRSPEC
Saturn Trigger - Epoch Two				
	14	SaturnMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(3) SATURN-TARGET
	15	MIRI Background	MIRI Medium Resolution Spectroscopy	(4) SATURN-BACKGROUND
	16	SaturnNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(8) SATURN-TARGET-NIRSPEC
Jupiter Trigger - Epoch One				
	2	JupiterMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(1) JUPITER-TARGET
	3	MIRI Background	MIRI Medium Resolution Spectroscopy	(2) JUPITER-BACKGROUND
	7	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(6) JUPITER-TARGET-NIRSPEC
	9	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(7) JUPITER-TARGET-LOC2-NIRSPEC

ABSTRACT

We propose a non-disruptive target of opportunity proposal to observe the aftermath of giant impact events or storm plumes on Jupiter and Saturn. Although such events are likely to occur within the foreseen lifetime of JWST, they remain impossible to predict in advance. JWST could provide spectroscopic maps of reflected sunlight (NIRSPEC) and thermal emission (MIRI), allowing us to derive the 3D structure of temperatures, winds, chemical composition and aerosols associated with these unique and rare events. For planetary impacts, the distribution of energy and chemicals (like ammonia) from the impact can reveal the impactor direction, penetration, and tensile strength; the reducing or oxidising chemistry can reveal the water content of the impactor; and the silicate debris can reveal its origins and chemical make-up. For giant planet storms, the thermal structure can reveal the energy released by the storm and the vigour of the rising motions; while the presence of unusual chemicals and aerosols can reveal the processes shaping jovian meteorology and provide a unique window onto the chemistry of the deeper, sub-cloud regions. The ToO will specifically target spectroscopy that cannot be reliably obtained from Earth, and will be supported by ground-based observations tracking the evolving impact/storm morphology. Only large impacts with early signs of debris, or significant and rare storms (particularly on Saturn), will be considered for this JWST trigger.

OBSERVING DESCRIPTION

Target of Opportunity Proposal to observe impact debris or significant storm plumes on Jupiter and Saturn.

JWST Proposal 1424 (Created: Monday, September 27, 2021 at 8:00:21 AM Eastern Standard Time) - Overview

* Observations have been designed for both planets, although the time request is based on the larger of the two requests. Note that this request is for a SINGLE trigger of EITHER Jupiter OR Saturn, so the total time request is smaller than that given in APT. Specifically:

> Jupiter Trigger One: 5.0 hours (with smart accounting applied).

> Jupiter Follow up: 5.0 hours.

> Saturn Trigger One: 3.8 hours (with smart accounting applied).

> Saturn Follow up: 3.8 hours.

Thus the total time requested is 10.0 hours (to cover the Jupiter trigger). Both planets have two epochs defined, in the hope that the impact/storm can be acquired twice within 7-14 days.

NOTES:

1. NIRSPEC POINTINGS: As the NIRSPEC FOV is the smallest of the modes proposed here, the Jupiter trigger features two NIRSPEC observations ~10 degrees longitude apart, to ensure that we capture the full extent of any impact scar or plume outbreak. This is not necessary for the smaller angular size of Saturn.

2. DITHERING: All observations use a 4-point dither for an extended source, but given that Jupiter and Saturn are rotating, software is being developed to allow us to properly coadd the sequence of dithers and integrations.

3. LOCATION: A dummy location has been specified for the purposes of submission, precise targetting will be provided at the time of the trigger. Note that we request the impact/storm location to be observed close to the central meridian, so a negative source-target separation constraint has been used to ensure that the target is nowhere near the planetary limb.

4. BACKGROUND: A single MIRI background frame (an offset to sky, 90" away from Jupiter or Saturn) should be acquired to characterise any anomalous slopes or contributions. This should be executed as near as possible to the science frames. Given the brightness of Jupiter and Saturn, we do not foresee the need for background observations with NIRSPEC.

5. GROUPS: As some regions of Jupiter's spectrum (particularly those beyond 11 μm) are likely to saturate in the second group, we have set $\text{ngroups}=5$ (the minimum possible, with the expectation that the final group will be saturated and discarded) and used sufficient iterations to build up our signal-to-noise. For the brightest regions, this will allow us to subsample the exposure time to create sharper images (Jupiter's rotation will cause blur during the science exposures).

Proposal 1424 - Targets - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	JUPITER-TARGET	STD=JUPITER	TYPE=PGRAPHIC, LONG=0, LAT=0	
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES				
	(2)	JUPITER-BACKGROUND	STD=JUPITER	TYPE=POS_ANGLE, RAD=90, ANG=0, REF=NORTH	
	Comments: Sky background for MIRI observations of Jupiter. Extended=YES				
	(3)	SATURN-TARGET	STD=SATURN	TYPE=PGRAPHIC, LONG=0, LAT=45	
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES				
	(4)	SATURN-BACKGROUND	STD=SATURN	TYPE=POS_ANGLE, RAD=60, ANG=0, REF=NORTH	
	Comments: Sky background for MIRI observations of Saturn. Extended=YES				
	(6)	JUPITER-TARGET-NIRSPEC	STD=JUPITER	TYPE=PGRAPHIC, LONG=0, LAT=0	
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES					
(7)	JUPITER-TARGET-LOC2-NIRSPEC	STD=JUPITER	TYPE=PGRAPHIC, LONG=10, LAT=0		
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES					
(8)	SATURN-TARGET-NIRSPEC	STD=SATURN	TYPE=PGRAPHIC, LONG=0, LAT=45		
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES					

Proposal 1424 - Observation 10 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 10: JupiterMIRI-Trigger												Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI Background (Obs 11)]												
	Comments: To ensure that the MIRI observation takes place when the longitude is close to the central meridian, we have specified a 'Target Window' ±17 degrees either side of the longitude of interest. However, the precise central longitude will not be known until ~2 months ahead of the observations.												
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(1)	JUPITER-TARGET	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			
	F1000W	ALL				NO				FULL			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	

Proposal 1424 - Observation 10 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	10 After 2 by 7 Days to 14 Days
	Sequence Observations 10, 11, 12, 13, Non-interruptible
	DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET FROM JWST LESS THAN 0.03
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET CALLISTO FROM JWST GREATER THAN 10"
	SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6"

Proposal 1424 - Observation 11 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 11: MIRI Background												Mon Sep 27 13:00:21 GMT 2021	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [JupiterMIRI-Trigger (Obs 10)]													
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Solar System Targets	#	Name			Level 1			Level 2			Level 3			
	(2)	JUPITER-BACKGROUND			STD=JUPITER			TYPE=POS_ANGLE,RAD=90,ANG=0,REF=NORTH						
	Comments: Sky background for MIRI observations of Jupiter.													
	Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel					Simultaneous Imaging				Imager Subarray			
		ALL					NO				FULL			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		

Proposal 1424 - Observation 11 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 10, 11, 12, 13, Non-interruptible
	DEFAULT WINDOW: ANGULAR RATE JUPITER-BACKGROUND FROM JWST LESS THAN 0.03
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-BACKGROUND BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND CALLISTO FROM JWST GREATER THAN 10"

Proposal 1424 - Observation 12 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 12: JupiterNIRSPEC-Trigger											Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Need additional solar system target window for this observation: SEPARATION BETWEEN (4 JUPITER-GRS) AND JUPITER FROM JWST LESS THAN -5.7"											
This constraint was formerly OK but does not compute in APT 25.4.2. Justification: Observations should be done with the target (GRS) not just visible, but also close enough to disk center. Tony Roman helped us determine this SEPARATION constraint would make sure the GRS is at a low emission angle when observed.												
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(6)	JUPITER-TARGET-NIRSPEC	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method											
	NONE											
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	

Proposal 1424 - Observation 12 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 10, 11, 12, 13, Non-interruptible
	SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6"
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-NIRSPEC BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC CALLISTO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-NIRSPEC FROM JWST LESS THAN 0.03

Proposal 1424 - Observation 13 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 13: JupiterNIRSPEC-Trigger											Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Need additional solar system target window for this observation: SEPARATION BETWEEN (4 JUPITER-GRS) AND JUPITER FROM JWST LESS THAN -5.7" This constraint was formerly OK but does not compute in APT 25.4.2. Justification: Observations should be done with the target (GRS) not just visible, but also close enough to disk center. Tony Roman helped us determine this SEPARATION constraint would make sure the GRS is at a low emission angle when observed.											
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(7)	JUPITER-TARGET-LOC2-NIRSPEC	STD=JUPITER				TYPE=PGRAPHIC, LONG=10, LAT=0					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method											
	NONE											
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	

Proposal 1424 - Observation 13 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 10, 11, 12, 13, Non-interruptible
	SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC JUPITER FROM JWST LESS THAN -6"
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-LOC2-NIRSPEC BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC CALLISTO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-LOC2-NIRSPEC FROM JWST LESS THAN 0.03

Proposal 1424 - Observation 5 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 5: SaturnMIRI-Trigger												Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning Observing Template: MIRI Medium Resolution Spectroscopy Background Observations:[MIRI Background (Obs 6)] <i>Comments: The targetted longitude can take any value - there is no requirement for a specific longitude, provided it is observed as close as possible to the central meridian.</i>												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(3)	SATURN-TARGET	STD=SATURN				TYPE=PGRAPHIC, LONG=0, LAT=45						
	<i>Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian.</i> Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			
	F1000W	ALL				NO				FULL			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	

Proposal 1424 - Observation 5 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	On Hold Awaiting triggering due to planetary impact or eruption of a substantial storm plume. Target Of Opportunity response time 14 Days 14 After 5 by 7 Days to 14 Days Sequence Observations 5, 6, 8, Non-interruptible DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET FROM JWST LESS THAN 0.03 DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-TARGET RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-TARGET TITAN FROM JWST GREATER THAN 10" SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -5"
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Proposal 1424 - Observation 6 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 6: MIRI Background												Mon Sep 27 13:00:21 GMT 2021	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [SaturnMIRI-Trigger (Obs 5)]													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Solar System Targets	#	Name			Level 1			Level 2			Level 3			
	(4)	SATURN-BACKGROUND			STD=SATURN			TYPE=POS_ANGLE,RAD=60,ANG=0,REF=NORTH						
	Comments: Sky background for MIRI observations of Saturn.													
	Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray				
	F1000W	ALL				NO				FULL				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		

Proposal 1424 - Observation 6 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 5, 6, 8, Non-interruptible DEFAULT WINDOW: ANGULAR RATE SATURN-BACKGROUND FROM JWST LESS THAN 0.03 DEFAULT WINDOW: NOT OCCULTATION OF SATURN-BACKGROUND BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-BACKGROUND BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND TITAN FROM JWST GREATER THAN 10"
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Proposal 1424 - Observation 8 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 8: SaturnNIRSPEC-Trigger											Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Need additional solar system target window for this observation: SEPARATION BETWEEN (4 JUPITER-GRS) AND JUPITER FROM JWST LESS THAN -5.7" This constraint was formerly OK but does not compute in APT 25.4.2. Justification: Observations should be done with the target (GRS) not just visible, but also close enough to disk center. Tony Roman helped us determine this SEPARATION constraint would make sure the GRS is at a low emission angle when observed.											
Diagnosics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(8)	SATURN-TARGET-NIRSPEC	STD=SATURN				TYPE=PGRAPHIC, LONG=0, LAT=45					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method											
	NONE											
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	2	5	false	true	NONE	4	20	644.206	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	

Proposal 1424 - Observation 8 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 5, 6, 8, Non-interruptible SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -5" DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET-NIRSPEC BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET-NIRSPEC BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC TITAN FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET-NIRSPEC FROM JWST LESS THAN 0.03
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Proposal 1424 - Observation 14 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 14: SaturnMIRI-Trigger												Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI Background (Obs 15)]												
Diagnostics	Comments: The targetted longitude can take any value - there is no requirement for a specific longitude, provided it is observed as close as possible to the central meridian.												
	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	Solar System Targets	#	Name	Level 1				Level 2				Level 3	
		(3)	SATURN-TARGET	STD=SATURN				TYPE=PGRAPHIC, LONG=0, LAT=45					
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian.													
Extended=YES													
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			
	F1000W	ALL				NO				FULL			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	

Proposal 1424 - Observation 14 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	14 After 5 by 7 Days to 14 Days Sequence Observations 14, 15, 16, Non-interruptible DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET FROM JWST LESS THAN 0.03 DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-TARGET RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-TARGET TITAN FROM JWST GREATER THAN 10" SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -5"
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Proposal 1424 - Observation 15 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 15: MIRI Background												Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [SaturnMIRI-Trigger (Obs 14)]												
Diagnostics	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(4)	SATURN-BACKGROUND	STD=SATURN				TYPE=POS_ANGLE,RAD=60,ANG=0,REF=NORTH						
	Comments: Sky background for MIRI observations of Saturn.												
	Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			
	F1000W	ALL				NO				FULL			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427	

Proposal 1424 - Observation 15 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 14, 15, 16, Non-interruptible DEFAULT WINDOW: ANGULAR RATE SATURN-BACKGROUND FROM JWST LESS THAN 0.03 DEFAULT WINDOW: NOT OCCULTATION OF SATURN-BACKGROUND BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-BACKGROUND BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND TITAN FROM JWST GREATER THAN 10"
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Proposal 1424 - Observation 16 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 16: SaturnNIRSPEC-Trigger											Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Need additional solar system target window for this observation: SEPARATION BETWEEN (4 JUPITER-GRS) AND JUPITER FROM JWST LESS THAN -5.7" This constraint was formerly OK but does not compute in APT 25.4.2. Justification: Observations should be done with the target (GRS) not just visible, but also close enough to disk center. Tony Roman helped us determine this SEPARATION constraint would make sure the GRS is at a low emission angle when observed.											
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(8)	SATURN-TARGET-NIRSPEC	STD=SATURN				TYPE=PGRAPHIC, LONG=0, LAT=45					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method											
	NONE											
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	2	5	false	true	NONE	4	20	644.206	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	

Proposal 1424 - Observation 16 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 14, 15, 16, Non-interruptible SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -5" DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET-NIRSPEC BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET-NIRSPEC BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC TITAN FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET-NIRSPEC FROM JWST LESS THAN 0.03
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Proposal 1424 - Observation 2 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 2: JupiterMIRI-Trigger												Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI Background (Obs 3)]												
	Comments: To ensure that the MIRI observation takes place when the longitude is close to the central meridian, we have specified a 'Target Window' ±17 degrees either side of the longitude of interest. However, the precise central longitude will not be known until ~2 months ahead of the observations.												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(1)	JUPITER-TARGET	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			
	F1000W	ALL				NO				FULL			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	

Proposal 1424 - Observation 2 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	On Hold Awaiting triggering due to planetary impact or eruption of a substantial storm plume. Target Of Opportunity response time 14 Days 10 After 2 by 7 Days to 14 Days Sequence Observations 2, 3, 7, 9, Non-interruptible DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET FROM JWST LESS THAN 0.03 DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET CALLISTO FROM JWST GREATER THAN 10" SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6"
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Proposal 1424 - Observation 3 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 3: MIRI Background												Mon Sep 27 13:00:21 GMT 2021	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [JupiterMIRI-Trigger (Obs 2)]													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Solar System Targets	#	Name			Level 1			Level 2			Level 3			
	(2)	JUPITER-BACKGROUND			STD=JUPITER			TYPE=POS_ANGLE,RAD=90,ANG=0,REF=NORTH						
	Comments: Sky background for MIRI observations of Jupiter. Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel					Simultaneous Imaging				Imager Subarray			
		ALL					NO				FULL			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		

Proposal 1424 - Observation 3 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 2, 3, 7, 9, Non-interruptible
	DEFAULT WINDOW: ANGULAR RATE JUPITER-BACKGROUND FROM JWST LESS THAN 0.03
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-BACKGROUND BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND CALLISTO FROM JWST GREATER THAN 10"

Proposal 1424 - Observation 7 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 7: JupiterNIRSPEC-Trigger											Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Need additional solar system target window for this observation: SEPARATION BETWEEN (4 JUPITER-GRS) AND JUPITER FROM JWST LESS THAN -5.7" This constraint was formerly OK but does not compute in APT 25.4.2. Justification: Observations should be done with the target (GRS) not just visible, but also close enough to disk center. Tony Roman helped us determine this SEPARATION constraint would make sure the GRS is at a low emission angle when observed.											
Diagnos	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(6)	JUPITER-TARGET-NIRSPEC	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method											
	NONE											
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	

Proposal 1424 - Observation 7 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 2, 3, 7, 9, Non-interruptible
	SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6"
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-NIRSPEC BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC CALLISTO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-NIRSPEC FROM JWST LESS THAN 0.03

Proposal 1424 - Observation 9 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Observation	Proposal 1424, Observation 9: JupiterNIRSPEC-Trigger											Mon Sep 27 13:00:21 GMT 2021
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Need additional solar system target window for this observation: SEPARATION BETWEEN (4 JUPITER-GRS) AND JUPITER FROM JWST LESS THAN -5.7" This constraint was formerly OK but does not compute in APT 25.4.2. Justification: Observations should be done with the target (GRS) not just visible, but also close enough to disk center. Tony Roman helped us determine this SEPARATION constraint would make sure the GRS is at a low emission angle when observed.											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Solar System Targets	#	Name	Level 1			Level 2			Level 3			
	(7)	JUPITER-TARGET-LOC2-NIRSPEC	STD=JUPITER			TYPE=PGRAPHIC, LONG=10, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method											
	NONE											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	

Proposal 1424 - Observation 9 - Extreme Events on Gas Giants: Impacts and Storm Eruptions

Special Requirements	Sequence Observations 2, 3, 7, 9, Non-interruptible
	SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC JUPITER FROM JWST LESS THAN -6"
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-LOC2-NIRSPEC BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC CALLISTO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-LOC2-NIRSPEC FROM JWST LESS THAN 0.03